

Problems: Categorical and Numeric Variables

Q1. [red, blue, blue, red, green, grey]

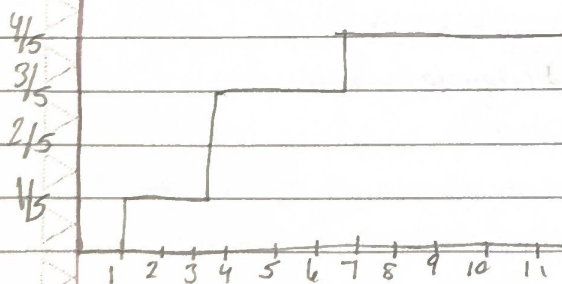
a. represent as matrix of one-hot encoded variables

	Red	Blue	Green	Grey
1	1	0	0	0
2	0	1	0	0
3	0	1	0	0
4	1	0	0	0
5	0	0	1	0
6	0	0	0	1

b. Sample proportions for each label.

Label	Sample Proportion
Red	$\frac{1}{3}$
Blue	$\frac{1}{3}$
Green	$\frac{1}{6}$
Gray	$\frac{1}{6}$

c. Consider $X = [1, 3, 3, 6, 11]$. Compute + sketch empirical CDF for this sample.



Q2. a. Suppose random variable X and transform it as $Y = a + bX$ w/ $b > 0$.

Dist of X given by $F_X(x)$, Dist of Y , $F_Y(y)$? What is the density, $f_Y(y)$?

• Def of CDF: $F_Y(y) = P(Y \leq y)$, $Y = a + bX$, so $F_Y(y) = P(a + bX \leq y)$

• $a + bX \leq y \rightarrow bX \leq y - a \rightarrow X \leq \frac{y-a}{b}$, so $F_Y(y) = P(X \leq \frac{y-a}{b})$

• $P(X \leq \text{something})$ is the def of F_X evaluated at something, so $F_Y(y) = F_X(\frac{y-a}{b})$

$$f(y) = \frac{d}{dy} F(y) \quad f_Y(y) = \frac{d}{dy} F_X\left(\frac{y-a}{b}\right) = f_X\left(\frac{y-a}{b}\right) \cdot \frac{d}{dy} \left(\frac{y-a}{b}\right)$$

$$f_Y(y) = \frac{1}{b} f_X\left(\frac{y-a}{b}\right)$$

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b. $X = \{1, 3, 4\}$. Mean?

$$\frac{1+3+4}{3} = \left(\frac{8}{3}\right)$$

c. Square all values. Mean of X^2 ?

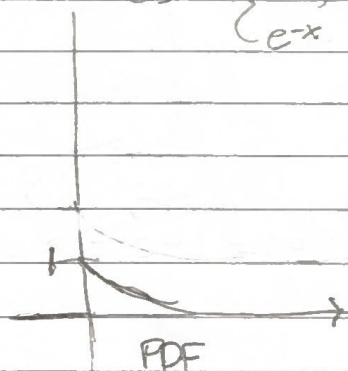
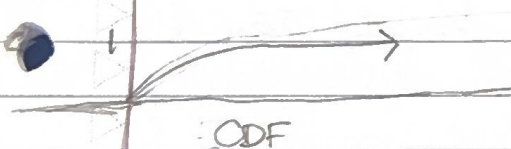
$$\frac{1^2+3^2+4^2}{3} = \frac{1+9+16}{3} = \left(\frac{26}{3}\right)$$

d. Compute square of mean from part b. Does this = mean of square from part c?

$$\left(\frac{8}{3}\right)^2 = \frac{64}{9} \neq \frac{26}{3}$$

$$3. F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} 0, & x < 0 \\ e^{-x}, & x \geq 0 \end{cases}$$



b. probability $X \geq 3$? Probability $X \leq 35$?

$F(x) = P(X \leq x)$ and $P(X \geq x)$ are complementary

$$P(X \geq 3) = 1 - P(X < 3) = 1 - F(3) = 1 - (1 - e^{-3}) = e^{-3} \approx 0.0498$$

$\approx 5\%$ chance that X is 3 or more.

$$P(X \leq 35) = F(35) = 1 - e^{-35} \approx 1. \text{ Almost a } 100\% \text{ chance that } X \text{ is } 35 \text{ or less.}$$

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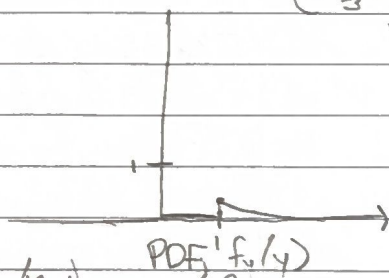
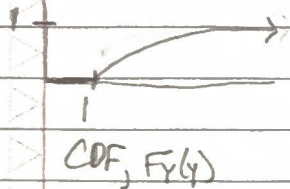
Q3. C. Transform X by $Y = 1 + 3X$. What is the distribution of Y , $F_Y(y) = \Pr[Y \leq y]$

$$F_Y(y) = F_X\left(\frac{y-1}{3}\right) \rightarrow F_Y(y) = \begin{cases} 0, & \frac{y-1}{3} < 0 \\ 1 - e^{-\frac{y-1}{3}}, & \frac{y-1}{3} \geq 0 \end{cases}$$

$$\downarrow F_Y(y) = \begin{cases} 0, & y < 1 \\ 1 - e^{-\frac{y-1}{3}}, & y \geq 1 \end{cases}$$

$$-\frac{1}{3}(y-1)$$

$$f(y) = \begin{cases} 0, & y < 1 \\ \frac{e^{-\frac{y-1}{3}}}{3}, & y \geq 1 \end{cases}$$



$P(Y < 10) = F(10) = 1 - e^{-\frac{10-1}{3}} = 1 - e^{-3} \approx .950$. Approximately 95% chance Y is < 10 .

Q4. a. $F(x) = \frac{1}{1+e^{-x}} = (1+e^{-x})^{-1}$

$$u = 1 + e^{-x}$$

$$u' = -e^{-x}$$

$$-1 \cdot u^{-2} \frac{du}{dx} = \frac{-1}{u^2} \frac{du}{dx}$$

$$f(x) = \frac{1}{(1+e^{-x})^2} \cdot -e^{-x} = \frac{e^{-x}}{(1+e^{-x})^2}$$

$F(x)$

$f(x)$

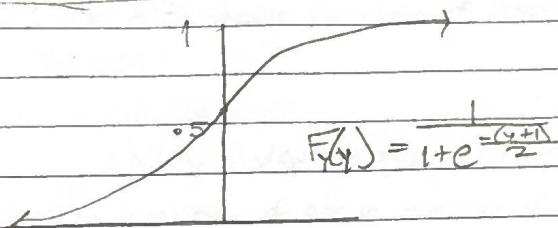
b. $P[X \geq .8] = 1 - P[X \leq .8] = 1 - F(.8) \approx 1 - .69 \approx .31$

$P[X \leq .3] = F(.3) \approx .57$

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c. Transform X by $Y = 2X - 1$. Dist of Y ?

$$F\left(\frac{y+1}{2}\right) = 1 + e^{-\frac{y+1}{2}}$$



$$P[Y < 0] = F\left(\frac{1+e^{-1/2}}{2}\right) \approx 0.62$$

Q5 a. median of exponential distribution?

$$1 - e^{-x} = 0.5 \rightarrow e^{-x} = 0.5 \rightarrow -x = \ln(0.5) \rightarrow x = \ln 2 \approx 0.693$$

b. median of logistic distribution?

$$\frac{1}{1+e^{-x}} = 0.5 \rightarrow 1 = 0.5(1+e^{-x}) \rightarrow 2 = 1+e^{-x} \rightarrow 1 = e^{-x} \rightarrow -x = \ln(1) \rightarrow x = 0$$

c. Quantile function of exponential distribution?

$$F(x) = 1 - e^{-x} = u \quad e^{-x} = 1 - u \rightarrow \ln(e^{-x}) = \ln(1 - u) \rightarrow x = -\ln(1 - u)$$

$$F^{-1}(u) = -\ln(1 - u)$$

d. Quantile function of logistic distribution?

$$F(x) = \frac{1}{1+e^{-x}} = u \quad 1+e^{-x} = \frac{1}{u}$$

$$e^{-x} = \frac{1}{u} - 1$$

$$\ln\left(\frac{1}{u} - 1\right) = -x$$

$$x = -\ln\left(\frac{1}{u} - 1\right)$$

$$F^{-1}(u) = -\ln\left(\frac{1}{u} - 1\right)$$